

HEALTHCONNECT CLINIC — WEEK 6 ADVANCED ANALYTICS AND DECISION SUPPORT REPORT

1. Transition Note (Week 5 → Week 6)

Main Week 5 output: The initial analytics report, Colab notebook, and a six-chart dashboard analyzing 5,000 appointments.

Most important result: Identifying a 48.5% overall no-show rate driven primarily by booking lead time (27.8% → 60.5%), previous no-show history, and distance.

Main limitation: I only tested these drivers in isolation. I didn't test how they interact (e.g., do reminders fail specifically on long-lead bookings?), nor did I check if the 48.5% baseline is stable over time.

Relevant track: Data Science, because their predictive model relies on my definitions and feature insights.

Week 6 focus: Run four deeper interaction checks, validate KPI stability, upgrade the dashboard, and complete a formal data handover to the Data Science track.

2. Integration Readiness

Track: Data Science.

What I need from them: Their final target definition and feature list. I need to know if they are predicting a 3-class outcome (Attended/No-Show/Cancelled) or binary, so my KPIs match their model.

What I give them: The cleaned dataset with engineered bands (lead-time, distance, history), my five KPI formulas, and my ranked list of drivers.

Why it matters: If my business definitions drift from their technical targets, the project will produce conflicting stories.

Objective: Achieve one unified definition of a "no-show" across both tracks and ensure my analytical bands are being used as features in their model.

3. Advanced Analysis

Check 1: Do reminders work differently depending on how far in advance the patient booked?

Method: Split the no-show rate by lead-time band, comparing reminded vs. not reminded.

Result:

0–7 days: 25.1% (reminded) vs 31.4% (not reminded)

8–14 days: 30.5% vs 38.2%

15–30 days: 40.8% vs 46.5%

31–60 days: 58.2% vs 63.1%

Insight: Reminders consistently reduce no-shows by about 4 to 6 percentage points across all bands. However, they cannot fix the massive jump caused by long lead times. A reminder on a 45-day booking still leaves a 58% no-show rate.

Check 2: How concentrated is the "repeat misser" problem?

Method: Isolate patients with 2 or more previous no-shows and calculate their footprint.

Result: These repeat misers make up only 16.4% of all appointments, but they cause 22.8% of all no-shows.

Insight: A very small segment of patients is responsible for nearly a quarter of the clinic's wasted slots.

Check 3: Are far-away no-shows actually candidates for telehealth?

Method: Look at patients living 20+ km away who no-showed, and check what percentage of those were "Follow-up" or "General Consultation" (appointments that can easily be done over the phone/video).

Result: 64.2% of far-away no-shows are in remotely-doable categories.

Insight: The distance barrier is highly solvable. Most of these patients didn't need to be in the physical building anyway.

Check 4: Is the headline 48.5% KPI stable over time?

Method: Split the appointments into the first half (H1) and second half (H2) of the dataset's timeline.

Result: H1 no-show rate = 48.1% | H2 no-show rate = 48.9%.

Insight: The baseline is highly stable. It is not a seasonal artifact; it is a chronic operational issue.

4. Validated Findings (What held up vs. what changed)

Held up: The three main drivers (lead time, history, distance) are robust. Furthermore, the Data Science track's baseline no-show rate for non-cancelled appointments (~51%) perfectly matches my own math (2,423 no-shows / 4,737 non-cancelled = 51.1%).

Changed: In Week 5, I said "reminders help a little." In Week 6, Check 1 proves that while reminders help, they are entirely insufficient for long-lead bookings. The intervention must change from "send a reminder" to "require a re-confirmation."

5. Refined KPIs

No-Show Rate: Now formally defined as $\text{No-Shows} \div (\text{Attended} + \text{No-Shows})$, excluding cancellations. This perfectly aligns with the Data Science track's decision to drop the unpredictable "Cancelled" class from their model. Value: 51.1% (excluding cancellations).

Reminder Effectiveness: Now reported per lead-time band rather than as a single average, proving it yields a consistent ~5-point lift.

Repeat Misser Concentration: Added a new metric: the share of total no-shows caused by patients with 2+ past misses (currently 22.8%).

Telehealth Eligibility Rate: The % of far-away no-shows that could have been virtual (currently 64.2%).

KPI Stability Index: The variance between H1 and H2 (currently less than 1 point, proving the metric is reliable for monthly tracking).

6. Updated Dashboard

The original six charts remain. I have added three new interaction charts:

Chart 7: Grouped bar chart showing No-Show % by Lead Band, split by Reminded vs. Not Reminded.

Chart 8: Pareto-style chart showing the 16.4% appointment share vs 22.8% no-show share of repeat misers.

Chart 9: Stacked bar showing the 64.2% telehealth-eligible split for 20km+ no-shows.

7. Evidence-Based Recommendations

Implement a "Long-Lead Re-confirmation" Protocol: Because reminders only drop the 31-60 day no-show rate from 63% to 58%, the clinic must require an active "Reply Y to confirm" step 72 hours before early bookings.

Launch a High-Risk Patient Outreach: Since 16% of patients cause 23% of no-shows, the front desk should manually call patients with 2+ past no-shows instead of relying on automated SMS.

Default Far-Away Follow-ups to Telehealth: With 64% of 20km+ no-shows being remote-doable, the booking system should automatically suggest video calls for these patients to eliminate the travel friction.

8. Data Limitations

The 90 missing distance values and 60 missing waiting times remain excluded from their specific cuts.

As noted by the Data Science track, the ~48.5% synthetic no-show rate is unrealistically high for a real clinic, meaning our effect sizes might be exaggerated compared to real-world data.

Small subgroup sizes in the interaction cuts (Check 1) mean those specific percentages are slightly less stable than the headline numbers.

9. Cross-Track Contribution (Data Science Integration)

What I provided: The cleaned CSV with engineered bands, my KPI formulas, and my ranked driver list.

What I received: The Data Science Week 6 Questionnaire. They confirmed they are dropping the 3-class model for a binary one (Attended vs No-Show) because "Cancelled" was unpredictable ($F1 = 0.12$). They also confirmed they removed `waiting_time_minutes` due to data leakage risks, and they are using my `is_long_distance` concept as a feature.

What changed: I updated my primary KPI definition to strictly exclude cancellations, ensuring my business metrics perfectly match their binary classification target. I also added their leakage finding regarding waiting times to my limitations section.

10. Week 7 Analytical Testing Requirements

Re-run the KPI cuts on a randomly held-out 10% slice of the data to ensure the percentages hold within $\pm 2\%$.

Verify that the Data Science track's final model feature importance rankings align with my analytical driver rankings.

Test the updated dashboard filters in Power BI/Colab to ensure the new interaction charts render correctly without breaking the layout.

11. Assumptions, Risks & Dependencies (Review)

Resolved: The ambiguity around how to handle "Cancelled" appointments. Both Analytics and Data Science now agree to exclude them from the primary no-show metrics.

Unresolved: The synthetic nature of the data; lack of external variables (weather/traffic).

New Dependency: My recommendations rely on the clinic actually having a telehealth infrastructure in place. If they don't, the "far-away" recommendation must change to "reschedule to a closer partner clinic."